4-1 Development of Information Technology

Point!

1 Advancements in Information Technology and Future Society

(1) Flow of Societal Development

Name	Explanation	Impactful Technologies and Tools
([1]Hunter-gatherer society)	• From the birth of humanity • A society where people live through hunting.	Stone axes, bows and arrows, fire-starting techniques, language, etc.
([2]Agrarian society)	• From around 13,000 BCE • A society where people lead a stable life by growing and harvesting food.	Agricultural tools, irrigation, etc.
([3]Industrialized society)	• From the mid-18th century ([4]post-industrial revolution) • A society where industrialization advanced through mass production using machinery, facilitated by the invention of the steam engine and electricity.	Steam engines, industrial machinery, printing technology, etc.
([5]Information society)	• From the latter half of the 20th century • A society where automation has been promoted by the advancement of computers.	Computers, the internet, etc.

(2) ([6]Society 5.0) : A new society where virtual space (cyberspace) and real space (physical space) are integrated to solve various social issues and achieve economic development.

2 New Information Technologies Attracting Attention

(1) ([7]Big Data) : Vast collections of data in various forms, with different characteristics and of various types. Data consists of three elements: data ([8]volume), data ([9]variety), and data ([10]generation/update frequency).

(2) ([11]AI (Artificial Intelligence)) : A technology that uses computers to artificially mimic human intellectual behavior.
<Examples> Autonomous driving, telemedicine, smart agriculture, smart homes, etc.

(3) ([12]Machine Learning) : A technology where computers learn from large amounts of data to extract rules and patterns from the data.

(4) ([13]IoT) : Technology that supports daily life by connecting household appliances and other items to the internet, thereby allowing the appliances to communicate with each other.

- ([14]Smart Home) : A residence that utilizes IoT and AI technologies to provide a more comfortable living environment.

(5) ([15]VR (Virtual Reality)) : A technology that allows users to experience an illusion of reality in a simulated environment.

(6) ([16]AR (Augmented Reality)) : A technology that uses computer processing to enhance the real-world environment perceived by human senses (vision, etc.).

3 Changes in Life Caused by Information Technology

(1) ([17]**Technostress**) : The overall term for mental and physical problems that arise from using computers.
- ([18]**Technophobia**) : A state of anxiety or fear towards using computers due to a sense of discomfort or lack of confidence in handling them.
- ([19]**Techno-Addiction**) : Excessive dependency on computers.
- ([20]**VDT Disorder**) : Shoulder stiffness and eye strain caused by prolonged computer use.

(2) ([21]**Digital Divide**) : The gap between those who can utilize information technology and those who cannot.

Warm Up

Answer the following questions.

(1) Choose the one that is the most appropriate sequence of societal development from the options A to **D** below, and answer using the letter.

A Hunter-gatherer society → Agrarian society → Industrialized society → Information society
B Agrarian society → Hunter-gatherer society → Information society → Industrialized society
C Agrarian society → Hunter-gatherer society → Industrialized society → Information society
D Hunter-gatherer society → Agrarian society → Information society → Industrialized society

(2) Choose the most closely related option for the following sentences from the options **1)** to **5)** below, and answer using the letters.

[1] Technology that connects all things (objects) to the internet.
[2] A technology in which machines analyze data to extract rules and patterns from the data.
[3] A vast amount of data that is difficult for humans to fully comprehend.
[4] A virtual space which is created by a computer and allows for experiences that feel like reality.

A Machine Learning **B** Big data **C** IoT (Internet of Things)
D Virtual Reality **E** Augmented Reality

(3) What is the term for the gap between those who can utilize information technology, such as the internet, and those who cannot?

Explanation

(1) **A**
(2) [1] **C** [2] **A** [3] **B** [4] **D**
(3) Digital Divide

Try

1 Read the following passage and answer the questions.

In ancient times, people lived by hunting animals and gathering plants. In such a ([1]) society, technologies like stone axes, bows and arrows, fire-starting techniques, and even language itself supported the community. Eventually, humanity settled on fertile land and led a life growing crops. In this kind of ([2]) society, the community was supported by technologies such as the invention of writing, agricultural tools, astronomy, and irrigation.

From the mid-18th century, when the steam engine was invented in Britain, the ([3]) Revolution occurred, leading to the formation of ([4]) society. Economic activities were stimulated by operating industrial machinery in large factories and by the mass production, transportation, and consumption of goods. Natural sciences and printing technologies supported ([4]) society.

Around the middle of the 20th century, with the development of media such as radio and television and the invention of the computer, the ([5]) society emerged. Humankind is currently aiming for a human-centered (A)new information society that harmonizes economic development with the resolution of social issues through systems that highly integrate cyberspace (virtual space) and physical space (real space).

(1) Choose the terms that best fit into the blanks [1] to [5] in the passage from the options **A** to **F**, and answer using the letters.

A Industrial **B** Hunter-gatherer **C** Information
D The internet **E** Agrarian **F** Industrialized

(2) What is the term for a society like the one referred to by the underlined section A?

2 Answer the following questions.

(1) Choose the most closely related options for the following sentences from the options **A** to **F** below, and answer using the letters.

[1] Vast amounts of data collected and accumulated from people, objects, society, and nature, which is then analyzed and utilized for problem-solving, thanks to advancements in information technology.
[2] An environment where all objects are connected to the internet and communicate with each other.
[3] Technology that enhances reality by using computers to add information to real-world environments.
[4] Residences that utilize IoT and AI technologies to achieve a more comfortable living environment.

A AI **B** Augmented Reality **C** Virtual Reality
D IoT (Internet of Things) **E** Smart Home **F** Big Data

(2) Choose the one that is not a characteristic of Big Data from the options **A** to **D**, and answer using the letter.

A The data has low versatility **B** The amount of data is enormous
C The types of data are diverse **D** The data has a high frequency of occurrence

3 Choose the term that best fits into the blank in the following text from the options **A** to **D** below, and answer using the letter.

The disparity that arises between those who can use information and communication technologies like the internet and computers and those who cannot is referred to as ( ).

A Technostress **B** Digital Divide
C VDT Disorder **D** Techno-Addiction

Exercise

1 Complete the following table by filling in the blanks [1] to [4] with the appropriate terms.

Name	Explanation
([1]) society	• From the birth of humanity • A society where people live through hunting.
([2]) society	• From around 13,000 BCE • A society where people lead a stable life by growing and harvesting food.
([3]) society	• From the mid-18th century (post-industrial revolution) • A society where industrialization advanced through mass production using machinery, facilitated by the invention of the steam engine and electricity.
([4]) society	• From the latter half of the 20th century • A society where automation has been promoted by the advancement of computers.

2 Answer the following questions.

(1) Choose the most closely related options for the following sentences from the options **A** to **D** below, and answer using the letters.

[1] An environment where all objects are connected to the internet and communicate with each other.

[2] A technology that uses computers to artificially mimic human intellectual behavior.

[3] Technology for constructing virtual environments that can be perceived by humans.

[4] A technology that adds information (still images, videos, etc.) by using computers to real-world environments.

A Virtual Reality **B** IoT (Internet of Things) **C** Augmented Reality **D** AI

(2) Choose one statement that is most inappropriate in regard to Big Data analysis from the options **A** to **D** below, and answer using the letter.

A Doctors are able to make quick and accurate diagnoses by consolidating information such as treatment progress and test results of patients.

B Monitor real-time traffic conditions, predict traffic congestion, and suggest optimal routes.

C Analyze consumer behavior patterns based on the results of group interviews.

D Analyze the large volume of posts on social media to identify trending keywords and understand current trends.

3 Choose the term that best fits into the blank in the following text from the options **A** to **D** below, and answer using the letter.

Anxiety and frustration from being unable to adapt to the operation of information devices such as computers or smartphones, as well as excessive dependency on the devices, are collectively referred to as ( ).

A Technostress **B** Digital Divide **C** VDT Disorder **D** Techno-Addiction